CMSC 27100-3 Due: Friday, October 22, 2021

Problem Set 3

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1 Number Theory

Problem 1: Suppose that $GCD(a, b, c) = 1$ where GCD is extended to three integers in the natural way. Are there necessarily two numbers from $\{a, b, c\}$ that are relatively prime? Give a proof.

Problem 2: Define $-[a]_m = [-a]_m$ where $m \neq 0$. Is this negation a valid function? Give a proof.

Problem 3: Suppose $a \equiv_m b$ where m is a composite number greater than 2. Demonstrate that there are 3 distinct values of m such that $a \bmod m = b \bmod m$.

2 Induction

Problem 4: Prove the commutativity of addition using only the axioms of Peano arithmetic. You may want to state and prove an intermediate lemma.